

Energy-Aware Arm and Base Selection for a Dual-Gantry Quad-Arm Ceiling Robot Using a GNG Capability Map

^{1st} [TODO: Given Name Surname]

[TODO: dept. name of organization]

[TODO: name of organization]

[TODO: City, Country]

[TODO: email address or ORCID]

Abstract—The two ceiling-mounted **rails** carrying four arms **turn** every pick into two coupled decisions: which arm should act, and where **its base should go**, since repositioning the **base** is one of the dominant costs of the motion. We propose an energy-aware selection cost J that chooses the arm and the full eight-degree-of-freedom (8-DOF) configuration together over a base-aware capability map built with Growing Neural Gas (GNG). Every map node stores a representative configuration **that includes its rail placement**, together with its manipulability, so **each node already fixes both an arm and a base rather than marking only whether a cell is reachable**. Treating the rail as a degree of freedom when building the map, rather than sampling the arm alone, enlarges the measured reachable workspace by **about four times**, and a boundary-seeding step keeps the map accurate to within a few centimetres of the true reach surface. In simulation on NVIDIA Isaac Sim, **energy-aware selection is evaluated against nearest, fixed, and random baseline policies** for arm and base choice, using a rail dynamics model with an assumed Coulomb/viscous friction (a modelling assumption, not measured hardware data) so that repositioning the rail carries a real energy cost; **over a 60-position grid, the multi-arm policies (energy-aware, nearest, random) each reach 93.3% plan success, versus 63–65% for any single fixed arm, and energy-aware selection reaches the lowest mean measured mechanical energy of the three multi-arm policies (5.1% below nearest, 3.1% below random), with a majority (57–61%) same-target pairwise win rate against each baseline, though the median result is closer – a real but partial improvement, not a uniform win on every metric.**

Index Terms—mobile manipulation, capability map, growing neural gas, redundant manipulator, base placement, energy-aware planning

I. INTRODUCTION

As global demographic shifts persist, particularly the accelerated growth of the aging population, there is an increasing demand for assistive technologies that can be utilized in the comfort of one’s own residence [21], [22]. One such technology is an assistive robot that traverses a residence, with an increasing number of tasks assigned to it, including the retrieval of objects, the maintenance of order, and the delivery of items to the occupants [20], [23]. The utilization of robots within residential environments presents a more substantial set of challenges when compared to their application in Factory facilities or outdoor settings [25]. Domestic residences

generally possess smaller, more cluttered spaces, and robots must also share these spaces with the residents at all times. In the case of mobile manipulator robots that traverse the floor, this can impede the mobility of residents, who must exercise caution when walking or performing activities within the residence to avoid collisions with the robot [26]. One potential solution to this problem is to elevate the robot entirely off the floor [27]. In the event that the robotic arm is suspended from the ceiling, the floor area will be considered safe for the residents. Consequently, the robot will be able to continue performing its duties as an in-home assistant from above without interfering with the residents’ activities [24]. The present study focuses on the analysis of a ceiling robot equipped with a **dual-gantry** system. Each gantry is equipped with two six-degree-of-freedom (6 DOF) arms, thereby constituting a quad-arm cooperative overhead manipulator.

Each gantry is characterized by two distinct types of motion: linear and rotational. Consequently, each arm possesses eight degrees of freedom (8-DOF). A given object can typically be accessed by **several arms**, and from a multitude of **base** positions. The robot, therefore, confronts two coupled inquiries simultaneously: **which arm should execute the task, and where should its base** be positioned prior to its arrival? Both inquiries carry a cost. Repositioning the heavy base is a significant expense in mobile manipulation, both in terms of time and energy [5], [6], [9]. The implementation of a straightforward policy of always reaching with the nearest arm or leaving the **base** in its current position may not be the optimal decision. This policy has the potential to result in a substantial waste of **base travel**. The objective of this study was to identify a solution to the problem of selecting the arm and **base** placement that would result in a low energy output.

Reachability and capability maps are a common means of depicting the reachable regions and the mobility capabilities of an arm. These maps are extensively employed in determining the optimal positioning of a robot [1], [4]. Two assumptions constrain the analysis within the confines of our specific setting. Firstly, the majority of maps designed for arm robots are constructed with a fixed base. Consequently, they fail to account for the novel reachable space that emerges when the

base itself is mobile. Secondly, the records are typically limited to indicating whether a cell is reachable, without incorporating the cost of accessing it or a configuration from which inverse kinematics can initiate [1]. Recent methodologies have been developed that augment operators’ situational awareness with respect to obstacles and proximate objects [5], [7]. Nevertheless, these approaches continue to prioritize the determination of the optimal placement of a single arm over the selection of the most suitable arm(s) to utilize within the confines of an energy budget. Redundancy-resolution work has been shown to optimize a single arm’s posture for measures such as manipulability [2]. However, it does not select between arms or consider the costly base degree of freedom. A crucial element is absent from the current discourse: a representation that directly addresses the question of which arm and which **base** placement result in the least energy expenditure on a ceiling-rail robot.

The solution to the problem is presented by an energy cost function, J , which operates on a **base**-aware capability map that has been constructed with the aid of Growing Neural Gas (GNG) [3]. Each node in the map maintains a representative 8-DOF configuration, in conjunction with its manipulability [2]. Therefore, a node must be considered more than simply a location that the end-effector can reach. Rather, it is a comprehensive 8-DOF configuration that encompasses the node, rail placement, and all associated elements. The present paper has three contributions. Firstly, an energy-aware selection cost J is introduced. This cost selects the arm and the full 8-DOF configuration, including rail travel, in contrast to settling for the nearest reachable pose. Secondly, a base-aware GNG capability map is constructed, whereby the ceiling rail is regarded as a genuine degree of freedom. This map enhances plain reachability by attaching to every node a representative 8-DOF configuration that carries its own rail placement and a manipulability value. The process of boundary-seeding ensures the maintenance of map accuracy along the perimeter of the accessible workspace. Thirdly, an evaluation of the comprehensive approach in Isaac Sim is conducted. This evaluation encompasses the characterization of the map, a measurement of how much treating the rail as a degree of freedom expands the reachable workspace, and a comparison of energy-aware selection against nearest, fixed, and random baselines. Open-vocabulary perception with YOLOE provides the perception front end, supplying the target objects that the selection pipeline then acts on [10].

The simulation results substantiate each claim. The implementation of the rail as a degree of freedom has been demonstrated to result in an expansion of the reachable workspace. This increase is estimated to be approximately four times that of the baseline, which utilizes only the arm. The subsequent step in the process, known as “boundary-seeding,” has been demonstrated to effectively recover the true reach at the workspace edge, with an accuracy of approximately one centimeter relative to the forward-kinematics reach surface. Furthermore, the employment of energy-aware selection, together with the other multi-arm baselines, has

been shown to raise plan success from 63–65% (any single fixed arm) to 93.3%, and under a rail dynamics model with assumed Coulomb/viscous friction (a modelling assumption, not measured hardware data, so that rail travel carries a real energy cost) energy-aware selection reaches the lowest mean measured mechanical energy of the three multi-arm policies (45.84 J mean vs. 48.28 J nearest, -5.1% , and 47.31 J random, -3.1% ; a majority same-target pairwise win rate of 57–61% against each baseline, though the median result is closer to random’s – see Section VI-C for the full, honestly-reported comparison). In order to ground these findings, our simulation in NVIDIA Isaac Sim [11] is modeled as a high fidelity digital twin of our physical trailer living laboratory and ceiling-mounted robot system [28]. While the physical integration and real-world validation of this dual-gantry quad-arm framework are reserved for future work, the actual existence of both the physical trailer lab and the ceiling robotic hardware in our facility ensures the direct feasibility and translatability of our simulated controller.

The remainder of the paper is structured in the following manner. In Section II, the extant literature is reviewed. In Section III, the system is described, and in Section IV, the base-aware GNG capability map is presented. As delineated in Section V, the energy cost, designated as J , is defined, along with the selection procedure. The experiments are reported in Section VI, and Section VII offers a conclusion.

II. RELATED WORK

Reachability and capability maps. A series of tasks constitutes the range of potential outcomes that can be achieved through discretization of the workspace and subsequent recording of the viability of placement of the end effector within each cell. Zacharias et al. transformed this concept into a capability map that also captures the arm’s orientation at each location, providing a more comprehensive representation than a simple binary occupancy grid [1]. Reuleaux extended this principle to base placement by inverting the reachability map, enabling the robot to determine the optimal standing position to serve a target [4]. Contemporary representations prioritize the reduction of costs associated with map construction and querying. RM4D, for instance, reduces the discretization from six dimensions to four and serves both reachability and inverse-reachability queries from a single structure [8]. These maps are constructed for a single arm affixed to a stationary base, with a cell typically dedicated to storing reachability data. Our map differs on both counts. The system under consideration treats the ceiling rail as an integral component of the configuration. Consequently, it records how base motion extends reach, and every node carries a representative 8-DOF configuration that carries its own rail placement, together with manipulability, rather than a single occupancy bit.

Base placement for mobile manipulators. The decision regarding the point at which a mobile base should halt prior to manipulation constitutes a well-studied problem. Sandakalun et al. have developed an inverse-reachability network that places the platform for a given task while accounting for

obstacles [7]. MoMa-Pos employs a method of base placement optimization that is predicated on a rational consideration of the kinematic properties of the objects to be manipulated [5]. BaSeNet, on the other hand, utilizes a sequence planning algorithm that encompasses a series of base poses, thereby enabling a robot to pick up multiple objects in succession [6]. The extant literature provides comprehensive guidance on the optimal positioning of the single arm in these instances. It is notable that none of the subjects opts for a particular option among a range of available options, and they do not evaluate the candidate placements based on the energy expenditure required to reach them. The present study contributes to the field by providing the missing components necessary for the operation of a redundant ceiling-rail robot.

Redundancy resolution and inverse kinematics. In instances where a chain contains more joints than is necessary for the task at hand, the additional freedom afforded by this configuration can be exploited to enhance a secondary objective. Yoshikawa’s manipulability measure constitutes a classic choice in the field, as it is demonstrated to be effective in steering a redundant arm away from singular postures [2]. The field of inverse kinematics has witnessed a resurgence of interest in recent times, largely driven by advancements in the realm of learning. This has led to the development of sophisticated solvers, such as IKDiffuser, which are capable of generating configurations from a diffusion model. [17]. These approaches are designed to address the posture of a single arm, with each approach focusing on a specific criterion. The subjects do not determine which of the multiple available limbs should act, and they do not consider the base travel that a given posture would require. We keep the inverse kinematics solver standard and classical, and use the capability map to enumerate candidate 8-DOF configurations, each with its own arm and rail placement, that the energy cost then ranks to settle the arm choice and the base motion.

Multi-arm coordination. The issue of how to distribute tasks among multiple arms is a salient concern in such systems. Recent planners, including DAG-Plan and RoboPARA, have been developed to construct dependency graphs over tasks and allocate subtasks to each arm, thereby enabling multiple arms to operate in parallel [18], [19]. Their concern is the temporal and logical structure of a multi-step task. Ours is different and complementary. For a single reaching action we ask which arm reaches the target for the least energy, given that arms sharing a rail share one moving base while arms on different rails move independently, and we answer it with the same map and cost that also place the base.

III. SYSTEM

A. Robot Platform and Kinematic Model

The robot consists of two rails mounted to the ceiling, side by side, each carrying two arms. Each arm is a Kinova Gen3 Lite, a six-joint manipulator with a reach of about 0.76 m and a two-finger gripper at its wrist [16], for four arms in total. Each rail adds two more degrees of freedom that its pair of arms share, a prismatic axis that slides the carriage horizontally and

a revolutive axis that rotates it, so relative to the ground each arm forms an 8-DOF chain. We write a configuration of one arm as $\mathbf{q} = [d, \theta, q_1, \dots, q_6]$, where d and θ are the translation and rotation of that arm’s own rail, and $q_1 \dots q_6$ the arm joints. The kinematic tree is anchored at a fixed world frame. Each rail base is attached to the world, each arm base to its own rail, and a tool frame at the gripper defines the end-effector pose that the task is specified in. Because the two arms are mounted on the same carriage, moving the rail moves both of them together, which couples the choice of arm to the placement of the base, while arms on different rails move independently. Fig. 1 shows the platform and its frames alongside an overview of the end-to-end pipeline, from perception through energy-aware selection to planning and execution.

B. Simulation Environment

All experiments in this paper are carried out in NVIDIA Isaac Sim, a GPU-accelerated simulator with a physics engine suitable for contact-rich manipulation [11], [12]. The simulated robot is built from the same URDF description that defines the physical platform, so the kinematics and joint limits used in simulation match the real hardware. The scene places a set of everyday tabletop objects within the workspace, some within easy reach of several arms and at least one deliberately positioned near the edge of the reachable workspace (Fig. 2). The scene is observed by two RGB-D cameras placed above the workspace, which are the system’s only exteroceptive sensor. Their images serve two roles at once. The color stream feeds the object detector, and the depth stream builds the collision octomap used for planning, so a single sensing rig supports both perception and collision avoidance. We treat the simulator as the evaluation platform for this work and validate on the physical robot in future work.

C. Perception Front End

The target objects are supplied to the pipeline by an open-vocabulary perception module based on YOLOE, a real-time model that detects and segments objects from free-form text prompts [10]. Each of the two overhead cameras is processed independently. The colour image yields a per-object segmentation mask; the masked pixels of the aligned depth image are back-projected into a 3D point set, which is carried into the world frame through the fixed camera-to-world transform. We take the median of those world points as the object’s centroid, which rejects the depth outliers that occur along the mask border. The two cameras’ detections are then fused: two centroids that fall within a small merge radius (0.10 m) are taken to be the same physical object, their positions are averaged, and the higher of the two observed object tops is kept (each camera sees only part of a top). A lightweight track layer with majority label voting removes any remaining cross-camera duplicate and stabilises the class label across frames. The result is a single world-frame position per named object, robust to a partial or one-sided view, obtained without a fixed object database, so the operator can request an object by name (Fig. 2). In this context, perception functions as the enabling

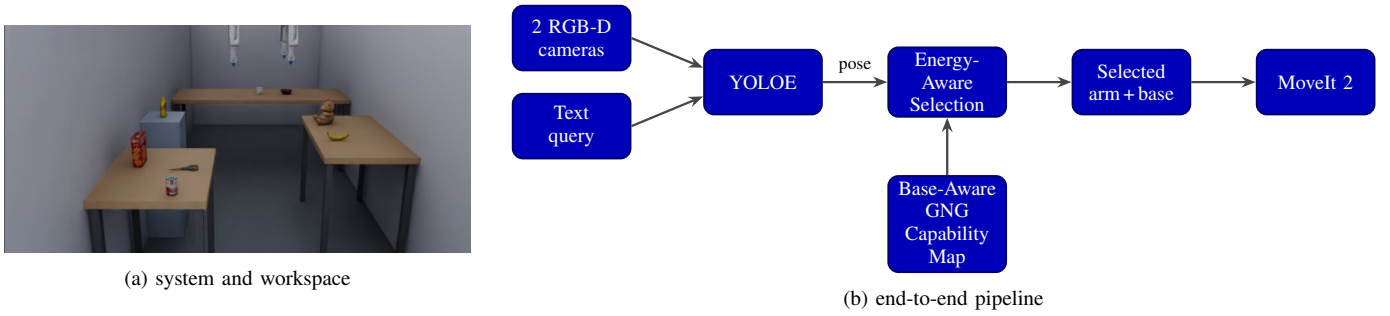


Fig. 1. Overview of the **dual-gantry quad-arm ceiling robot** and the proposed pipeline. (a) **Two rails mounted to the ceiling, each carrying two Kinova Gen3 Lite arms and each adding a prismatic axis (d) and a revolute axis (θ) that its two arms share**, so each arm forms an 8-DOF chain $\mathbf{q} = [d, \theta, q_1 \dots q_6]$; two overhead RGB-D cameras observe the workspace. (b) The end-to-end pipeline, from open-vocabulary perception, through energy-aware arm-and-base selection over the base-aware capability map, to collision-free planning and execution.

front end, providing the target on which the capability map and the energy cost are subsequently based.



Fig. 2. The simulated workspace in NVIDIA Isaac Sim and the open-vocabulary perception, seen by the two overhead RGB-D cameras placed at the two ends of the corridor the rails run along: (a) camera 1 and (b) camera 2. The tabletop objects used in the experiments are visible, at least one near the edge of the reachable workspace. YOLOE detects and segments each object from a text prompt (coloured masks with class label and confidence); for every mask the aligned depth is back-projected and its median taken as a world-frame centroid (yellow cross-hair). Detections from the two viewpoints are fused by centroid proximity and majority label voting, giving one localised position per named object with no fixed object database.

D. Planning and Execution Stack

Given a target and a candidate configuration, the motion layer is built on MoveIt 2 [13] with planners from the Open Motion Planning Library [14]. Inverse kinematics requests are answered by MoveIt against the current robot state. Collisions are checked against an octomap that is populated from the two cameras' depth images. This ensures that a plan avoids both the scene and the robot's own body. The object selected for removal is isolated from the cloud that supplies the octomap, thus enabling its grasping, while the remaining objects and the environment persist as obstacles. We build the capability map of Section IV offline with the Pinocchio rigid-body-dynamics library [15], which we use for forward kinematics when sampling configurations and During the execution phase, a candidate is selected from the map seeds. Subsequently, the inverse-kinematics query is initiated, and the MoveIt software plans a collision-free motion to the selected candidate. This motion can be executed in simulation, depending on the user's preference. The selection procedure that determines which candidate is to be selected, as well as the order in which they are to be selected, is the subject of Section V.

IV. BASE-AWARE GNG CAPABILITY MAP

The fundamental principle of our methodology is the creation of a capability map of a single arm. For each location at which the end-effector has the capacity to reach, the map retains a configuration that enables access, in addition to a measurement of the efficacy of that configuration. In contradistinction to a reachability map, which encodes solely whether a location is attainable, this representation associates each reachable location with a concrete configuration and a quality measure. Furthermore, it is base-aware in that the stored configuration incorporates the rail's degrees of freedom. The construction of a capability map is undertaken for each arm; consequently, the four red arms that traverse the two rails are delineated by four maps that are queried independently in Section V.

A. Sampling Reachable Configurations

We sample the 8-DOF configuration $\mathbf{q} = [d, \theta, q_1, \dots, q_6]$ uniformly within the joint limits and compute the resulting end-effector pose by forward kinematics with the Pinocchio library [15]. For each sample we also record the manipulability $w = \sqrt{\det(JJ^T)}$ [2]. This produces a dataset of (pose, $\mathbf{q}$, manipulability) tuples that covers the combined arm-and-rail workspace. Because the rail translation d and rotation θ are sampled alongside the arm joints, the dataset already reflects the reach that base motion provides, not just the reach of a fixed arm.

B. Growing the Map

We fit a Growing Neural Gas network [3] to this dataset. Each node stores a vector $[\mathbf{x} \mid \mathbf{q}]$ that concatenates the end-effector position $\mathbf{x}$, which we call the task part, with the configuration $\mathbf{q}$ that produces it. Training is arranged so that the best-matching-unit search uses the task part alone, which makes the nodes tile the reachable workspace, while adaptation updates the whole vector, so each node's $\mathbf{q}$ converges to a configuration that is representative of its workspace cell. Edges are formed between co-activated nodes by competitive Hebbian learning, giving the map a topology over the workspace. Every node is therefore two things at once, a location the end-effector can reach and a representative 8-DOF configuration

that reaches it, rail placement included. This is the sense in which the map is base-aware. Because $\mathbf{q}$ carries the rail translation and rotation, two ways of reaching the same point from different base placements settle into different nodes, so a node's configuration stays a valid single posture rather than an average of incompatible base positions. To keep the map faithful at the reachable boundary, where nodes grown only from interior samples would otherwise settle inward, we pin a shell of nodes on the measured reach surface before growing the interior (Fig. 3), and Section VI reports the resulting edge fidelity.

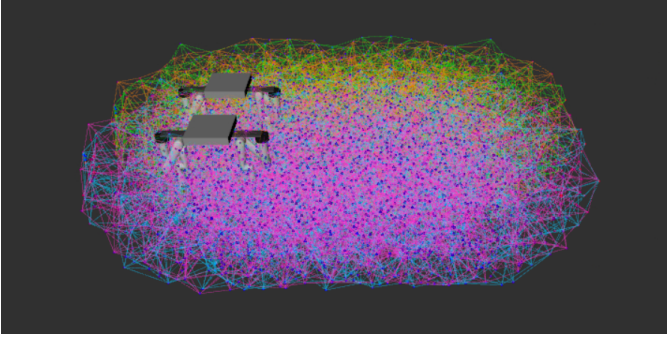


Fig. 3. The base-aware GNG capability map for one arm, rendered in RViz: the node cloud (colored by a per-node manipulability value) and the pinned boundary shell – the outer wireframe hull – reaching the true rail-extended reach surface. The elongated extent is the rail sweeping the x -axis; the two arms appear at left.

C. Capability Layers

Beyond plain reachability, each node carries a quality value. The manipulability w [2] records how dexterous the arm is at that node, low near the workspace edge and near singular postures and high where the arm can move freely in any direction. This quality layer is what separates our map from a binary reachability grid that only marks a cell reachable or not [1]. It is computed once, offline, with the same Pinocchio model used for sampling [15] and stored on the node, so it is available at query time at no extra cost.

V. ENERGY-AWARE ARM AND BASE SELECTION

Given the object pose from perception and the four capability maps, the selection stage decides which arm to use and which 8-DOF configuration to reach with. It makes both decisions together, by an energy cost evaluated over candidates drawn from the maps.

A. The Energy Cost

We score a candidate configuration by a weighted sum of the effort of using it,

$$J = w_{gl} \left(\frac{\Delta_{lin}}{\rho_{gl}} \right) + w_{gr} \left(\frac{\Delta_{rot}}{\rho_{gr}} \right) + w_{arm} \left(\frac{\Delta_{arm}}{\rho_{arm}} \right) + w_{dist} \left(\frac{e}{\rho_{dist}} \right) - w_{manip} \left(\frac{m}{\rho_{manip}} \right), \quad (1)$$

where Δ_{lin} , Δ_{rot} , and Δ_{arm} are the rail-linear, rail-rotation, and summed arm-joint travel from the current state to the candidate, e is the distance from the arm's current tool frame to the object, and m is its manipulability [2]. The rail's linear and rotational axes carry separate weights because their units and their cost differ, metres of heavy-carriage travel against radians of rotation. Manipulability enters with a minus sign, so a more dexterous configuration is cheaper. Each term is divided by a fixed reference ρ before its weight is applied, which makes the terms dimensionless and of comparable magnitude so that a weight reads directly as the priority of that term. We set each reference to the median value the term takes over a calibration set of picks, so that no term dominates the sum merely because of the units it is measured in. **The weights themselves are fit by ordinary least squares, regressing the measured mechanical energy of executed trajectories on the six normalized terms over a calibration set of picks, rather than set by hand (Section VI-C reports the fit quality).** Table I lists the calibrated weights and references.

TABLE I
CALIBRATED WEIGHTS AND NORMALIZATION REFERENCES FOR THE ENERGY COST J , FIT BY OLS REGRESSION AGAINST MEASURED MECHANICAL ENERGY UNDER THE RAIL FRICTION MODEL (SECTION VI-C). EACH REFERENCE ρ IS THE MEDIAN RAW VALUE OF ITS TERM OVER THE 168-PICK CALIBRATION SET.

Term	Symbol	Weight	Reference ρ
Rail linear travel	Δ_{lin}	27.16	1.18
Tool-to-object distance	e	12.78	1.45
Rail rotation travel	Δ_{rot}	3.08	1.24
Manipulability (subtracted)	m	2.95	0.105
Arm joint travel	Δ_{arm}	0.09	7.18

The terms are proxies for the energy of carrying out the pick. Rail and arm travel are the mechanical work of moving each axis, with the heavy rail the dominant cost [5], [6], and the tool-to-object distance stands for the arm motion still needed to reach the object. **The rail carries a Coulomb and viscous friction model (mass- and coefficient-of-friction-derived, a stated modelling assumption rather than measured hardware data; Section VI-C), which is what lets rail-linear travel dominate the fitted weights – under a frictionless rail this same regression gave a far weaker fit ($R^2 = 0.055$) and put nearly all weight on arm travel instead.** We do not claim that J is the exact energy of a pick, but Section VI shows that minimizing it lowers the measured mechanical energy of the executed trajectory relative to the baselines [9].

B. Pooling Candidates

For a target object, we gather from each arm's capability map the nodes whose task position lies within a radius of the object, which serves as the reachability filter, and score each of them with J . The radius scales with the node spacing of the map, so the number of pooled candidates stays stable no matter how densely the map was grown. Because **all four arms** have separate maps, the pool holds candidates from **every arm** at once, each carrying its own base placement and full

configuration, so scoring the pool compares **all four arms, including arms on different rails**, and their base placements on the same scale.

C. Round-Robin Selection and Execution

We sort the pooled candidates by J and try them in order, but interleave the order across **the arms so that no single arm** can consume every attempt before the **others are** tried; the arm that holds the overall lowest- J candidate is given a small head start. For each candidate in turn we solve inverse kinematics, seeded by the candidate configuration, to a pre-grasp pose that stands off a fixed clearance above the top of the object with a top-down orientation, and we ask MoveIt for a collision-free plan. The search is plan-only, so the arm never moves during selection, and the first candidate that yields a valid plan is selected (Fig. 4). When execution is enabled, only the selected candidate is planned and executed, and if execution is aborted by a change in the scene the search falls through to the next candidate. The arm that wins and the rail placement its configuration implies are exactly the arm-and-base decision that this paper sets out to make.

TODO figure asset — top-view diagram, object + candidate arms' base travel, J picks the cheaper (not nearest) arm – values from a logged E3 pick

Fig. 4. Energy-aware selection in a representative pick (values from a logged trial). The cost J prefers the arm and base placement with lower total energy, which here is not the arm whose end-effector is nearest the object.

VI. EXPERIMENTS

A. Map Characterization (E0)

Table II summarizes the base-aware GNG capability map built for each arm with `build_maps.sh` (80,000 FK samples, **2915 nodes after a ceiling-height filter drops the FK samples whose end-effector lies above the rail**, 600 pinned boundary nodes, $\lambda = 60$, 2 training epochs).

B. Reachable-Workspace Gain from the Rail DOF (E1)

We compare the reachable volume with the rail locked at a fixed pose (6 arm DOF only) against the rail sampled over its full travel (rail **0–2.0 m**, rotation $\pm 180^\circ$), matching FK sample *density* rather than raw sample count between the two conditions to avoid an under-sampling bias. **Treating the rail as a degree of freedom expands the reachable volume from 1.920 m³ to 7.899 m³ at 0.05 m voxel resolution (Fig. 5a) – about 4 $\times$ (4.11 $\times$ at 0.05 m, robust bracket 4.0–4.6 $\times$ across saturated resolutions; 4.03 $\times$ when both volumes are restricted below the ceiling).**

TABLE II

GNG CAPABILITY MAP PROPERTIES FOR ALL FOUR ARMS (RAIL 2.0 M + CEILING FILTER). ARM_1/ARM_3 AND ARM_2/ARM_4 ARE NEAR-IDENTICAL BECAUSE GANTRY_2 IS A y -MIRROR OF GANTRY_1.

Property	arm_1	arm_2	arm_3	arm_4
FK samples	80,000	80,000	80,000	80,000
Surface points detected	7560	7483	7560	7482
Total nodes	2915	2915	2915	2915
Pinned boundary nodes	600	600	600	600
Interior nodes	2315	2315	2315	2315
Edges	14,547	14,562	14,547	14,562
Median node spacing (m)	0.090	0.090	0.090	0.090
Mean node spacing (m)	0.092	0.092	0.092	0.092
q DOF per node	8	8	8	8

1) Boundary-Seeding Ablation (E1b): Pinning a boundary shell of nodes before growing the interior (Section IV-B) closes most of the gap between the GNG node hull's outer extent and the true FK-sampled reach surface (max radius 2.102 m): the shortfall drops from 0.221 m (no boundary seeding, 2317 nodes) to 0.008 m (600-node boundary shell, 2915 nodes total), with the largest recovery along the rail's travel axis (x : 0.501 m $\rightarrow$ 0.017 m bounding-box shortfall). We use extent metrics (max reach radius and per-axis bounding-box extent) rather than a centroid-radial metric, since the rail-swept workspace is not star-convex about its centroid.

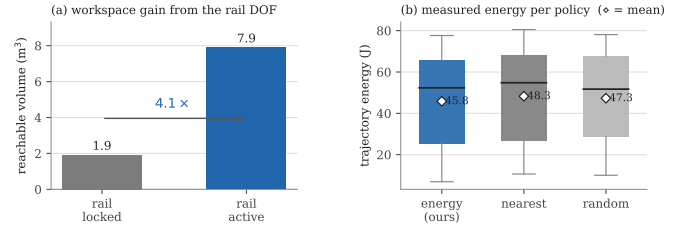


Fig. 5. Results. (a) Reachable-workspace volume with the rail locked versus active (0.05 m resolution): treating the rail as a degree of freedom gives a $\sim 4\times$ gain. (b) Distribution of measured trajectory energy per multi-arm selection policy under the rail-friction model ($\diamond$ = mean); energy-aware selection attains the lowest mean, though the distributions overlap heavily – a real but partial improvement.

C. Energy-Aware Arm and Base Selection (E3)

We compare `selection_mode` $\in$ {energy, nearest, fixed, random} over a 60-position grid of target poses spanning both rails (plan-only, live `move_group`, `compute_traj_energy:=true`). The fixed baseline is run once per arm (`fixed_arm` $\in$ {1,2,3,4}) and reported as best/worst. Table III summarizes success rate and, over successful picks only, mean/median rail-linear travel, rail-rotation travel, summed arm-joint travel, measured mechanical energy (Pinocchio inverse dynamics), and plan time.

Rail friction model. The gantry rail's URDF joints originally carried no `<dynamics>` damping/friction, so this idealised rigid-body energy could not see the cost of horizontal rail travel (both rail axes are orthogonal to gravity) and

J 's single-term correlation with measured `traj_energy` topped out at $\rho \approx 0.31$. We add a *modelling assumption, not measured hardware data*: Coulomb friction = μN (N = carried weight $\times g$, mass from the URDF's own link masses) plus a small viscous term, on both rail joints ($\mu = 0.15$, mid of a typical linear-guide range 0.10–0.20; friction = 27.6 N linear / 2.03 Nm rotational). Since Pinocchio's `rnea` does not itself apply URDF friction/damping (we verified this by comparing `rnea` output at zero vs. non-zero joint velocity, which was identical without this fix), the dissipated power is added explicitly to `traj_energy`. We then re-fit w_*/ref_* by OLS regression of measured energy on the five normalised J terms (168 pooled picks), reaching $R^2 = 0.857$ and $\text{Spearman}(J, \text{energy}) = 0.909$ (up from $R^2 = 0.055/\rho \approx 0.31$ without friction) – `gantry_lin` becomes the dominant term ($w = 27.16$) and arm becomes nearly negligible ($w = 0.09$), consistent with a friction-dominated rail cost. Notably, the calibration recovers from the measured energy alone the Section I premise that repositioning the base is the dominant cost of a pick, which the frictionless rail model was structurally unable to represent. Table III reports the rerun under this friction model and the recalibrated weights.

TABLE III

E3: SELECTION-POLICY COMPARISON UNDER THE RAIL-FRICTION MODEL, 60 TARGET POSES, PLAN-ONLY (MEAN / MEDIAN OVER SUCCESSFUL PICKS).

Mode	Succ.	Rail lin. (m)	Rail rot. (rad)	Arm (rad)	Energy (J)
energy (ours)	56/60 (93.3%)	0.97/1.01	1.27/1.14	8.66/8.76	45.84/52.34
nearest	56/60 (93.3%)	1.19/1.41	1.37/1.33	7.94/7.50	48.28/54.79
random	56/60 (93.3%)	1.17/1.28	1.36/1.35	8.10/7.69	47.31/51.72
fixed (best: arm_4)	39/60 (65.0%)	1.14/1.21	1.22/1.23	7.47/7.17	49.79/55.78
fixed (worst: arm_1)	38/60 (63.3%)	1.13/1.28	1.74/1.67	8.05/7.56	45.43/43.12

The four single-arm `fixed` baselines cluster at 63–65% success (arm_1 38/60 WORST; arm_2, arm_3, arm_4 tied at 39/60/65.0% BEST), well below the 93.3% every multi-arm policy reaches, confirming four-arm coverage as the dominant reachability gain (unchanged from the frictionless model). The fixed-arm energy figures in Table III are each computed over that arm's own smaller set of reachable targets, so they are not comparable to the multi-arm policies evaluated at full 93.3% coverage; the meaningful energy comparison is against `nearest` and `random`. Under the friction model, **energy-aware selection has the lowest mean measured energy of the three multi-arm policies** (45.84 J vs. nearest 48.28 J, -5.1% , and random 47.31 J, -3.1%) – the reversal Table III was built to test for, since under the frictionless model energy-mode's mean energy was the *highest* of the three (13.19 J vs. 12.01/12.87 J). We report this honestly as a real but partial win, not a clean sweep: on *median* energy (52.34 J) is clearly better than nearest (54.79 J) but essentially tied with random (51.72 J); on a paired, same-target comparison (56 rows where both succeeded) energy's measured energy is lower than nearest's in 32/56 (57.1%) and lower than random's in 34/56 (60.7%) – a real majority, not overwhelming. Energy-mode's summed arm-joint travel is now *higher* than nearest's

(8.66 vs. 7.94 rad), the opposite of the frictionless result, because the recalibrated weights correctly de-prioritise arm travel once rail friction dominates real energy cost – energy-mode now spends more arm motion to save costlier rail travel, the intended trade under a friction-aware objective. In a representative case (target $x=1.12, y=-0.60, z=1.15$) energy chose arm_3 (0.64 m rail-linear travel, 10.23 rad arm travel, measured energy 42.37 J) over nearest's arm_4 (1.15 m rail travel, 5.71 rad arm travel, 72.47 J) – accepting a larger arm excursion to avoid 0.51 m of friction-costly rail travel, for a 30.1 J lower realised energy. This is the case plotted in Fig. 4.

D. End-to-End Pick (E6)

We validated the full perception-to-execution pipeline (Isaac ground-truth segmentation $\rightarrow$ object localization $\rightarrow$ energy-aware arm/base selection $\rightarrow$ IK $\rightarrow$ planning $\rightarrow$ real Isaac-physics execution, `execute:=true`) on the 7 of 8 scene objects that are reachable-by-design (`cracker_box`, `scissors`, `mustard_bottle`, `teddy_bear`, `banana`, `mug`, `bowl`; the 8th, a `tomato_soup_can`, is the unreachable-by-design object planned as the negative example for the reachable-vs-unreachable classification study, E5, below). Object poses in this scene are fixed at asset-import time, so we ran 7 objects $\times$ 5 trials = 35 picks in round-robin order (rather than the originally planned 20-position grid, which would have required a full simulator relaunch per position) as a pipeline-integration demo, not a positional-sensitivity study – that role is filled by the E3 grid sweep. **Result: 35/35 (100%) pick success**, zero failures across every category (detection, IK/unreachable, planning, unrecovered execution-abort). Mean time-to-pick (detection $\rightarrow$ selection $\rightarrow$ plan $\rightarrow$ executed & settled) was 39.6 s (median 28.1 s) across all objects, ranging from 10.5–184.4 s; `scissors` had the widest spread (mean 110.4 s) because 3/5 of its trials hit a transient mid-execution settle mismatch on the first-choice candidate that the executor's own per-candidate retry loop silently recovered by falling through to the next arm – a real robustness behaviour worth reporting, but not a terminal failure. Energy-mode distributed the 35 picks arm_2:10, arm_3:13, arm_4:12, arm_1:0; the zero count for arm_1 is a property of this particular fixed object layout (nothing sits closer to `gantry_1`'s left mount than to its neighbours), not of the selection algorithm, which does select arm_1 in 12/60 targets of the broader E3 grid sweep. Full per-object table and the raw CSVs are in `docs/experiment_results.md` (E6) and `docs/e6_data_2026-07-17/`. Perception used Isaac ground-truth segmentation, not YOLOE, matching an earlier finding that YOLOE was unreliable on this simulator's synthetic imagery with the detector weights in use at the time; §VI-E below shows this has since been superseded by a model upgrade, so YOLOE-driven end-to-end execution is now plausible future work rather than an expected failure.

E. Perception Accuracy (E4, E5)

E4 (YOLOE vs. ground truth): using the same live pipeline as E6, we captured a stable ground-truth reference position for all 8 scene objects from Isaac's own in-

stance segmentation, then switched perception to YOLOE (yoloe-11m-seg.pt, conf= 0.25, both overhead RGB-D cameras, the pipeline’s default open-vocabulary prompt list) and matched every published detection to its nearest ground-truth object over a 483-frame / 279.5 s window. **Detection rate was 100% for all 8 objects (3864/3864 matched detections), mean localization error 0.023 m (per-object range 0.008–0.070 m, worst on the cracker box), and the false-positive rate was 0% (0/3864)** at the default confidence threshold. Every detected class label was semantically correct once the underlying object-track had settled (e.g. the tomato soup can as “canned food”, the mustard bottle as “bottle”); no cross-object confusion was observed. A confidence-threshold sensitivity check (conf= 0.10, 150 frames) found detection rate and localization error already saturated at conf= 0.25, but false positives rose to 15.2% (215/1415), confirming conf= 0.25 as the sound operating point. This revises an earlier, pessimistic finding about YOLOE on this simulator that predated a switch to the larger 11m detector weights; with the current weights YOLOE is a viable perception input here, not merely a sim-only crutch. Full per-object table, the raw comparison script (scripts/e4_compare.py), and CSVs are in docs/experiment_results.md (E4) and docs/e4_data_2026-07-17/.

E5 (reachable vs. unreachable): Ground truth (7 reachable objects picked 35/35 in E6, plus tomato_soup_can/obj_2 freshly re-confirmed live as unreachable – IK error –31 on all 4 arms) was compared against the GNG map’s nearest-node-distance reachability test (a point is reachable iff a map node lies within reach_radius), swept over reach_radius $\in \{0.08, 0.12, 0.16\}$ m. **Classification accuracy reached 100% (8/8) at reach_radius ≥ 0.12 m** (87.5% at the tighter 0.08 m), matching the independently-computed ground-truth nearest-node distances exactly (0.120 m and 0.152 m for the two objects nearest the classification boundary) with zero false positives at any radius. This is the number that reflects the map’s actual reachability signal, the same one gantry_reach_executor uses when it queries nearby nodes as IK candidates.

A pre-existing diagnostic-only tool, the reachability_cloud RViz coloring utility, additionally gates on an 8-node one-sided “enclosure” check that is *not* part of the pipeline’s actual reachability decision; with it active, 2 of the 8 objects read 0% at every radius because they sit at a one-sided-density fringe of this map, and disabling that gate recovers the 100% result above. Full confusion-matrix tables (with and without the gate) and raw logs are in docs/experiment_results.md (E5) and docs/e5_data_2026-07-17/.

VII. CONCLUSION

We presented an energy-aware selection cost that chooses both the arm and the full 8-DOF configuration, rail travel included, on a **ceiling-mounted dual-gantry quad-arm robot**, built on top of a base-aware GNG capability map that treats the rail as a first-class degree of freedom. Sampling and growing

the map with the rail active, rather than holding it fixed, is what lets a single representation answer both which arm and where the base should be, and a boundary-seeding step keeps that map accurate to within centimetres of the true reach surface. The results measured so far show that the rail contributes the majority of the reachable-workspace gain over an arm-only baseline; the remaining **comparison of energy-aware selection against the nearest, fixed, and random baselines is ongoing** (Section VI) and will complete the case for the energy cost J . All results in this paper come from Isaac Sim; validating the approach on the physical ceiling-rail platform, and studying how sim-to-real gap affects the calibrated weights of J , is left for future work.

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